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MIL-H-22881D(SHIPS)
13 October 1971
SUPERSEDING
MIL-H-22881C(SHIPS)
23 May 1969
(See 6.5)

MILITARY SPECIFICATION

HEATERS, WATER, STEAM HEATED,

NON-STORAGE TYPE

(COMPLETE WITH CONTROL ACCESSORIES)

1. SCOPE

1.1 Scope. This specification covers steam heated, continuous hot water generation type water heaters complete with controls and accessories for shipboard installation.

1.2 Classification.

1.2.1 Types. Heaters shall be of the following types and classes as specified (see 6.2):

Type I - Temperature feedback control
Type II - Feedforward control (quick-recovery type)

Class A - Personnel use (showers, lavatories, etc.)
Class B - Non-personnel use (dishwashers, laundry, etc.)

1.2.2 Capacity. Heaters shall be of the following minimum rated capacities, as specified (see 6.2 and 6.4):

Class A applications	Class B applications
10 gallons per minute (g.p.m.)	7.7 g.p.m.
20 g.p.m.	15.3 g.p.m.
30 g.p.m.	23.0 g.p.m.
40 g.p.m.	30.7 g.p.m.
60 g.p.m.	46.0 g.p.m.

The above capacity ratings shall be based on the continuous generation of 140°F. hot water from a supply water temperature of 40°F. (100°F. required temperature rise) for class A heaters and from 40° to 180°F. (140°F. required temperature rise) for class B heaters. The delay required to achieve this continuous and stable hot water generation rate, starting from a dead-ended condition (dead-ended for at least 1 hour), shall not exceed 40 seconds for a type I heater and 20 seconds for a type II heater.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
QQ-S-691 - Steel Plate, Carbon-Silicon, Carbon Molybdenum and Manganese-Molybdenum Alloys Hot Rolled, (Marine Boiler Quality).
PPP-P-40 - Packaging and Packing of Hand Tools.

MILITARY
MIL-P-116 - Preservation, Methods of.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-917 - Electric Power Equipment, Basic Requirements (Naval Shipboard Use).

FSC 4520

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MILITARY (continued)

MIL-D-1000 - Drawings, Engineering and Associated Lists.
MIL-D-1000/2 - Drawings, Engineering and Associated Lists.
MIL-F-1183 - Fittings, Tube, Cast Bronze, Silver Brazing.
MIL-V-13612 - Valves, Relief, Pressure and Temperature (For Hot Water Supply Systems).
MIL-P-15024 - Plates, Tags and Bands For Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.
MIL-M-15071 - Manuals, Technical: Equipments and Systems Content Requirements For.
MIL-I-17244 - Indicators, Temperature, Direct-Reading Bimetallic, (3 and 5 Inch Dial).
MIL-A-17472 - Asbestos Sheet, Compressed (Gasket Material).
MIL-V-19772 - Valves, Temperature Regulating, Automatic, Naval Shipboard.
MIL-F-20042 - Flanges, Pipe, Bronze (Silver Brazing).
MIL-S-24370 - Solenoids, Electrical (Naval Shipboard).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.
MIL-STD-278 - Fabrication Welding and Inspection, and Casting Inspection and Repair For Machinery, Piping and Pressure Vessels in Ships of the United States Navy.

DRAWING

MILITARY

810-1385917 - Thermometer Selection Guide.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A36 - Structural Steel.
A193 - Alloy-Steel Bolting Materials for High-Temperature Service.
A194 - Carbon and Alloy-Steel Nuts for Bolts for High-Pressure and High-Temperature Service.
A216 - Carbon-Steel Castings, Suitable for Fusion Welding for High-Temperature Service.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B16.5 - Steel Pipe Flanges and Flanged Fittings.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, New York 10018.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 First article inspection. Heaters furnished under this specification shall be products which have successfully passed first article inspection (see 4.2).

3.2 Materials. Materials used in the manufacture of water heaters shall be as specified in table I. This specification of materials is not intended to be restrictive, providing proposed alternate materials will give equal or better service than the materials specified. Naval Ship Engineering Center (NAVSEC) approval is required for the use of any alternate material.

Table I - Material list.

Part	Material
Water side of system (parts in contact with water) except coils, coil manifolds and non-metallic elements (see below)	Corrosion-resisting steel 90-10 copper nickel Copper silicon Copper Bronze
All parts on steam side of heater (in contact with steam or condensate) except coils and coil manifolds (see below)	Copper Carbon steel: QQ-S-691, ASTM A36, ASTM A216 Bronze 90-10 copper nickel Corrosion-resistant steel
Gaskets	Compressed asbestos sheet: MIL-A-17472 or commercial equivalent
O-rings and other non-metallic elements	Approved elastomers Polytetrafluoroethylene
Bolting	ASTM A193 grade B7, ASTM A194 grade 2H Corrosion-resistant steel Nickel-copper alloy Cadmium plated steel
Coils and coil manifolds	Copper Bronze 90-10 copper nickel Corrosion-resisting steel

3.2.1 Materials for non-magnetic heaters shall be in accordance with table I and shall have a magnetic permeability of not over 2.0 in the finished form, and shall be approved by the procuring activity.

3.3 Definitions. The following definitions are applicable to this specification:

(a) Temperature feedback control (type I). A control system where the outlet temperature is maintained under all load conditions by a temperature regulating valve which receives a feedback signal proportional to the outlet temperature. This feedback is compared with a reference setting and any error acts to re-position the temperature regulating valve to control steam flow to the heat exchanger in such a way as to reduce the error in outlet temperature.

(b) Feedforward control (type II). A control system where the outlet temperature is maintained under all load conditions by a pre-established relationship between the water flow demand, the steam pressure to the heat exchanger, and a blending of heated and unheated water. Does not require outlet temperature feedback for control.

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3.4 Design. Water heaters shall be of the steam activated type capable of heating and discharging water at the required temperature and capacity upon demand (see 3.6.1). Type I heaters shall flow the steam through coils heating element and water through the shell. Type II heaters shall flow the steam through the shell and water through the coils. The water discharged shall be uniformly heated, without stratification. A mixing chamber external to the heater to effect mixing of the heated water is acceptable. For strength purposes, the steam and water sides, and connections thereto, shall be designed for a working pressure of not less than 125 p.s.i.g. Heaters shall be designed to allow accessibility to all component parts requiring maintenance or repair. Heater control and regulating valves shall be of a self-contained design which does not rely on compressed air or other external power media for operation.

3.4.1 Controls and accessories (type I). Type I water heaters shall include the following controls and accessories:

(a) An automatic temperature regulating valve for controlling the flow of steam to the heat exchanger to maintain the required outlet water temperature. The temperature regulating valve shall conform to the design, material, and construction requirements of either type II or type III of MIL-V-19772. It shall be single seated, and shall provide consistent tight shutoff of steam flow to assure that the outlet water temperature does not rise above the limits specified in 3.6.2 during low load or no load conditions. Means shall be provided to secure the temperature setting of the valve from accidental change.

(b) A safety device (see note) utilizing a temperature sensing element independent of that used to operate the temperature regulating valve in (a) which will either directly shut off, or will act to shut off (see below), steam flow to the heat exchanger when the outlet water temperature rises 5° to 10°F. above the temperature regulation limit specified in 3.6.2. The safety device shall consist of an independent electrically operated, energize-to-open, single seated, tight shut-off type solenoid stop valve and shall be controlled by immersion type electric switch. Where a type II temperature regulator of MIL-V-19772 is provided for temperature control, the solenoid stop valve shall be installed directly in the supply steam line upstream of the temperature regulating valve so that when actuated to close (de-energized) it will shut off all steam supply to the heater. Where a type III temperature regulator of MIL-V-19772 is provided for temperature control, the solenoid stop valve may be as described above or may be installed in the steam pilot supply take-off to the regulating valve so that when actuated to close (de-energized) it will shut off steam supply to the pilot and thereby cause the regulating valve to shut off tightly. When the solenoid stop valve is actuated to close (de-energized) by an excessive outlet water temperature condition, a red tell-tale jeweled light, located at the solenoid valve, shall illuminate to indicate that steam supply has been shut off. Electrical requirements for the equipment shall be as specified in 3.4.4.

NOTE: Where it can be demonstrated that the heater provides adequate intrinsic safety, the special override safety device specified in 3.4.1(b) may be omitted. Written NAVSEC concurrence of the intrinsic safety of a particular heater design and approval for omission of these safety devices is required.

(c) Relief valve protection (see 3.5.11).
(d) Thermometer to indicate water outlet temperature (see 3.4.3 and 3.5.12).
(e) Automatic steam condensate drainage (see 3.4.3 and 3.4.5).
(f) Any filters and strainers necessary for protection of accessories such as regulating valves, steam traps, etc. (see 3.4.3).

3.4.2 Controls and accessories (type II). Type II water heaters shall utilize the water demand flow rate as the basis of control and shall automatically mix incoming water and heated water in a pre-set ratio to provide the rquired outlet temperature under all flow rates. Controls and accessories for type II heaters shall consist of the following:

(a) A 3-way mixing or blending valve which automatically mixes incoming water with water leaving the heat exchanger to produce outlet water at the required temperature. Means shall be provided to secure the temperature setting of the valve from accidental change.
(b) A self-contained steam regulating valve which automatically regulates steam pressure to the heat exchanger. This valve shall be loaded by water from the heater inlet and the set pressure shall be continuously adjusted as a function of the water demand flow rate. Design shall be such that steam supply is tightly shut off by the steam regulating valve when the water demand drops below 1/2 g.p.m.

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(c) A water loading valve which adjusts the loading pressure to the steam reducing valve as a function of the water demand flow rate. This valve shall also serve to provide thermal expansion protection for the water side of the heater.
(d) A pressure gage to indicate regulated steam pressure to the heat exchanger.
(e) A safety device (see note) actuated by an immersion type electric switch installed at the water outlet, which will either directly shut off, or will act to shut off (see below), steam flow to the heat exchanger when the outlet water temperature rises 5° to 10°F. above the temperature regulation limit specified in 3.6.2. The safety device shall consist of either a single-seated, energize-to-open solenoid stop valve as specified in 3.4.1(b) installed directly in the supply steam line upstream of the pressure regulating valve, or a 3-way, 2-position solenoid valve installed in the water loading line to the steam pressure regulating valve which will dump all loading pressure when de-energized and thereby cause the steam regulating valve to tightly shut off. When the solenoid valve is actuated to stop steam flow to the heat exchanger (de-energized) by an excessive outlet water temperature condition, a red tell-tale jeweled light, located at the solenoid valve, shall illuminate to indicate that steam supply has been shut off. Electrical requirements for the equipment shall be as specified in 3.4.4.

NOTE: Where it can be demonstrated that the heater provides adequate intrinsic safety, the special override safety device specified in 3.4.2(e) may be omitted. Written NAVSEC concurrence of the intrinsic safety of a particular heater design and approval for omission of these safety devices is required.

(f) Thermometer to indicate water outlet temperature (see 3.4.3 and 3.5.12).
(g) Automatic steam condensate drainage (see 3.4.3 and 3.4.5).
(h) Any filters and strainers necessary for protection of accessories such as regulating valves, steam traps, etc. (see 3.4.3).

3.4.3 Accessories. The thermometer, automatic condensate drainage components, and necessary filtration equipment shall be furnished either as part of the complete heater package by the manufacturer or may be furnished as separate items by the installing activity, as specified (see 6.2). Where these items are not supplied by the manufacturer as part of the heater package, the manufacturer shall provide all data necessary to assure installation of suitable and compatible items. This data shall include such information as required, operating characteristics, types, sizes, capacity, etc., and available sources of supply.

3.4.4 Electrical accessories. All electrical equipment shall conform to the applicable requirements of MIL-E-917. Solenoids shall conform to the requirements of MIL-S-24370. Solenoids shall be suitable for continuous duty operation in a 50°C. (122°F.) ambient and shall have either a watertight housing or be encapsulated or both. All electrical equipment shall be suitable for the power supply specified (see 6.2).

3.4.5 Drainage. The steam side of the water heater shall be provided with an automatic drainage system. Means shall be provided to easily manually drain both the steam and water sides of the heater.

3.4.6 Weights. The maximum allowable weights (wet) of water heaters shall be as shown in table II. Maximum weight shall include external chamber and other accessories, when furnished, but not the steam traps and filters or strainers.

Table II - Weight of water heaters (complete with accessories).

Rated capacity (g.p.m.)		Weight (pounds)
Class A	Class B	
10	7.7	350
20	15.3	490
30	23.0	615
40	30.7	710
60	46.0	750

3.4.7 Mechanical shock and vibration. All heaters shall be designed to meet the mechanical shock requirements of grade A, class I of MIL-S-901 and the vibration requirements of type I of MIL-STD-167.

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3.4.8 Inclining, pitching and rolling. The design of water heaters shall be such that they will perform satisfactorily when permanently inclined (30 degrees from the normal position in any plane).

3.5 Construction. The water heater shall be constructed and fabricated as specified in 3.5.1 through 3.5.12.

3.5.1 Shells. Shells shall be seamless castings, or consist of a sheet rolled into a cylindrical form and seam welded or forged into shape.

3.5.2 Flanges. Flanges shall be forged integral with the shell or made of plate, welded to the shell.

3.5.3 Shell liners. Shell liners, when used, shall be fabricated from sheet material either silver-brazed or welded. Liners and flanges for liners shall be assembled in such a manner that no portion of the shell is exposed to the liquid in the internal part of the completed shell. Liners shall fit securely inside the steel shell without wrinkling or buckling. Cladding of the shell is not an acceptable construction.

3.5.4 Heads or cover plates. Heads or cover plates shall be either cast or forged and shall be designed for bolting to the shell flange.

3.5.5 Heat transfer element. The heat transfer element may be either coils or U-tube bundle. If coils are furnished, they may be mounted on the heads or cover plates or may be mounted supported within the cylinder with suitable framework or brackets. The coils shall be so placed that, when steam is circulated inside the coils, full drainage will occur when the heater is in the upright position. No air traps shall interfere with circulation. The coils may be mounted in either a pancake or spiral nest. If U-tube heating element is furnished, the element, or bundle, shall be securely supported in a partial shell or wrapper extending into the storage tank. U-tube bundle shall be arranged to permit full drainage of the condensate.

3.5.6 End connections. End connections for all steam and water inlets and outlets shall terminate in take-down connections. The water inlet and outlet shall be supplied with unions in accordance with MIL-F-1183 (modified for O-ring and retaining ring seal). The steam supply and condensate drain connections shall be flanged, either with steel flat-faced flanges in accordance with ANSI B16.5 or bronze flanges in accordance with MIL-F-20042 depending on whether the pipes leading to and from the heat exchanger are steel or non-ferrous material. Each heater shall be equipped with a drain on the water side, that will permit complete drainage, and suitable cut-off valve. Threaded pipe connections shall not be used. Connections between components of the heater shall be as approved by NAVSEC.

3.5.7 The shell or head for type I heaters shall be equipped with bosses suitable for the steam control valve thermostat bulb, and relief valve. The thermostat control bulb retaining-sleeve shall be brazed or welded into the boss.

3.5.8 Welding and silver-brazing. Welding and silver-brazing shall be in accordance with MIL-STD-278.

3.5.9 Heater supports. Each heater shall be supplied with supports to elevate the heater to a height to permit all connections to be readily made under the heater.

3.5.10 Identification and information plates. Each heater shall have an identification plate in accordance with type C of MIL-P-15024 and MIL-P-15024/5, mounted on the shell of the heater and shall contain the following data:

(a) Steam pressure rating.
(b) Capacity in g.p.m.
(c) Manufacturer's name or trademark.
(d) Manufacturer's model or part number.

3.5.11 Pressure relief valve. Each type I heater shall have a non-isolatable pressure relief valve on the water side for thermal expansion protection. The relief valve shall be in accordance with type III of MIL-V-13612 and shall be provided with a ground joint union connection, with a tail piece suitable for brazing. The relief valve shall be set to relieve at 20 pounds per square inch (p.s.i.) (minus 0, plus 3 p.s.i.) above the shut off pressure of the supply water pumps (see 6.2).

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3.5.12 Thermometers. Each heater shall have a thermometer in the hot water outlet piping. Thermometers shall be 3 inch dial size, in accordance with MIL-I-17244 with a 20° - 240°F. range. Thermometer shall be so installed as to be readily visible to operating personnel and to be accessible for repair and replacement purposes. The thermometer well shall be suitable for socket welding or brazing in accordance with detail 12 of Drawing 810-1385917 (see 3.4.3).

3.6 Performance requirements.

3.6.1 Performance design parameters. The heater shall meet all specified performance requirements under the following design parameters, unless otherwise specified (see 6.2):

(a) Inclining, pitching, and rolling: as defined in 3.4.8.
(b) Supply water temperature: 40° - 60°F. (see 3.6.2).
(c) Supply water pressure: 40 ± 10 p.s.i.g.
(d) Supply steam temperature: saturation.
(e) Supply steam pressure: 40 ± 10 p.s.i.g.
(f) Water flow demand: zero to rated capacity (see 1.2.2).

3.6.2 Adjustable range. Class A heaters shall have an adjustable range of water discharge temperature settings of 100° to 140°F. Class B heaters shall have an adjustable range of water discharge temperature settings of 140° to 180°F. All heaters shall maintain the outlet water temperature within an accuracy of plus or minus 5°F. at any setting within this range and under any operating condition within the limits specified in 3.6.1. In order to meet the above, type II heaters may incorporate an adjustment to compensate for supply water temperature variations, however, the ratio of delivered water temperature change to inlet water temperature change shall not exceed 1:3 within the overall band specified in 3.6.1(b).

3.6.3 Response time. The time required to generate a continuous and stable full rate discharge of hot water from a completely dead-ended start (i.e. steam supply valve allowed to fully shut) shall not exceed 40 seconds for a type I heater and 20 seconds for a type II heater.

3.6.4 Temperature setting. Unless otherwise specified (see 6.2), class A heaters shall be supplied set for a discharge water temperature of 130°F.

3.7 Identification plates. Each heater shall have an identification plate(s) permanently attached to an exposed position. The identification plate(s) shall be made of a corrosion-resistant material and shall contain the following infformation, or a space therefor:

(a) Manufacturer's name.
(b) MIL-H-22881 and type, class and capacity.
(c) Inlet steam rating, if higher than specified herein.
(d) Manufacturer's model and part number or identification for entire heater package as well as for all major accessories or sub-components.
(e) Applicable manual number.
(f) Space for 9 digit CID number.

3.8 Drawings. Drawings shall be in accordance with categories A, G or H of MIL-D-1000 as applicable.

3.8.1 Preliminary drawings. Preliminary drawings which are sufficient to permit evaluation of the design and approval of materials, shall be submitted with bids to the procuring activity. These drawings shall show the following:

(a) Accurately scaled sectional assembly which clearly depicts the design and construction of the heater and all accessories or sub-components.
(b) Bill of materials listing specification, grade, condition, and any other data required to fully identify the properties of the materials proposed.
(c) Dimensions-overall, accessibility space, including disassembly clearances, and all other dimensions pertinent to installation including location of supports, bolt holes, size of bolts and location of welds if welding is used for attachment to foundation.
(d) Type, class, capacity and adjustable range of heater.
(e) Location, size and identifying data for all mechanical and electrical connections. Also recommended assembly torques, or equivalent procedures, for making up all joints and threaded assemblies within the heater package.
(f) Schematic diagram of electrical controls where such equipment is provided.

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(g) Weight (wet and dry) of heater and any limitations on installation such as orientation, length of straight pipe runs required, etc.
(h) Reference to any previous shock and vibration approval for the heater and test report numbers.
(i) Part numbers to permit replacement of all parts and sub-components.

3.8.1.1 If the supplier proposes to furnish equipment which previously has been supplied and for which the procuring activity has approved drawings on record, reference to the approved drawings will be acceptable in lieu of submitting preliminary drawings.

3.8.2 Final drawings. Final drawings and certification data sheets shall be submitted to the procuring activity for approval within 60 days after date of contract. These drawings shall be in accordance with types II and III of MIL-D-1000/2 except for extent of detail. Only the information required in 3.8.1 need be furnished for type II drawings. The following data, in addition to that required in MIL-D-1000/2, shall be furnished for the type III drawings:

(a) Ship identification.
(b) Applicable assembly drawing number(s).
(c) Applicable manual number.
(d) CID (APL) number.
(e) Adjustable range of heater.
(f) Type, class and capacity of heater.

3.8.3 Limited rights legend. When the Government has only limited rights in the data shown on the drawings, as determined by the contractual provisions regarding rights in technical data, the drawings furnished may be marked with the following restrictive legend:

"Furnished under United States Government Contract No._____. Shall not be either released outside the Government, or used, duplicated, or disclosed in whole or in part for manufacture or procurement, without the written permission of _____, except for: (a) emergency repairs or overhaul work by or for the Government, where the item or process concerned is not otherwise reasonably available to enable timely performance of the work; or (b) release to a foreign government, as the interests of the United States may require; provided that in either case the release, use, duplication or disclosure hereof shall be subject to the foregoing limitations. This legend shall be marked on any reproduction hereof in whole or in part."

3.9 Manuals. Manuals shall be furnished in accordance with type I of MIL-M-15071. The quantity and distribution of manuals shall be as specified (see 6.2). In addition to the requirements for type I of MIL-M-15071, the following shall be included as part of the manual contents:

(a) The approved engineering drawings for the heater (including certification data sheet). These drawings shall be supplemented by additional illustrations where necessary to adequately illustrate operation and maintenance. These additional illustrations may consist of blowouts, partial or full sections, etc., and may eliminate extraneous lines and details to clarify the interaction of parts.
(b) Table listing wrench sizes and assembly torques (or other equivalent procedures) for making up all joints and threaded assemblies.
(c) Instruction to permit overhaul by shipyard or other repair facility. These should include procedures for checking all critical dimensions subject to wear or change and the acceptable dimensional limits, surface finish condition, etc. Also the appropriate procedure (i.e. part replacement, correction at repair facility, or repair at manufacturer's facility) which should be followed to correct each case of damage or wear.
(d) Detailed disassembly and reassembly procedures. In addition to providing procedures for the complete disassembly and reassembly of the equipment, maintenance and troubleshooting sections shall contain, or refer to, only the limited disassembly and reassembly required to accomplish each particular operation. This is intended to reduce the possibility of unnecessary disassembly and unnecessary disturbance of adjustments when performing specific or limited maintenance or trouble shooting operations.
(e) Adjustment procedures.

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3.10 Special tools. Special tools for servicing the heater and associated valves shall be furnished in the quantity specified (see 6.2) or one set per heater when not specified. When practicable, brackets or other means of stowage shall be provided on the heater. Special tools are defined as those tools not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service (DCAS)).

3.11 Workmanship. Heaters shall be free from defects affecting their appearance or that may affect their operation. Mating surfaces of flanges and mounting pads or feet of supports shall be machined true. The workmanship will be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article tests. The first heater of a particular type, class and capacity furnished under a contract or order shall be subjected to the examination and tests specified in 4.2.1 through 4.2.5 unless that type, class and capacity heater has successfully passed previous first article tests. Documentation of applicable first article tests shall be provided.

4.2.1 Hydrostatic test. The heater shall be hydrostatically tested to 250 p.s.i.g. on both the steam and water sides. Full test pressure shall be applied to the water side with the steam side drained of water. The water side shall then be drained and the steam side pressurized to full test pressure. In each case the pressure shall be held for 5 minutes. Any evidence of leakage, permanent distortion, or collapse shall result in rejection of the heater. If desired by the manufacturer, the steam portion may be tested separately before final assembly, and then the entire assembled heater hydrostatically tested on the water side after assembly.

4.2.2 Operational tests. The heater shall be connected for test as required by the supplier's instructions. All gages, thermometers and regulator controls shall be calibrated before the test. The instrumentation shall register inlet and outlet pressure and temperature of the water, at rated flow, inlet pressure and temperature of the steam, and flow of water and steam to the heater or condensate discharge from the heater. These tests shall be conducted with inlet water at 40°F. and with water outlet temperature set at 140°F. for class A heaters and 180°F. for class B heaters. Where tests are conducted at a supply water temperature other than 40°F. (due to ambient conditions), suitable and approved re-calculation of capacities required shall be made to compensate, or the test may be conducted with the set temperature adjusted so as to provide the temperature rise required above.

4.2.2.1 Full load recovery test. The time required, under full load conditions to establish a continuous and stable generation of heated water (140°F. for class A heaters and 180°F. for class B heaters) from a fully dead-ended start (dead-ended for at least one hour) shall be recorded and shall be in accordance with 3.6.3. Duration of full load test shall be 5 minutes.

4.2.2.2 Intermittent load test. The heater shall be tested in accordance with table III. All data required by table III shall be recorded and the heater shall meet all performance requirements specified herein at all times during the test.

Table III - Intermittent load test.

Flow rate percent capacity (approximate)	Duration of test (minimum minutes)	Water					Steam			
		G.P.M.	Gallons	Inlet temperature	Outlet temperature	Temperature rise °F.	Temperature	Pressure	Pounds/hr	Pounds
10	5									
20	5									
0	10									
30	2									
# 0	60									
50	2									
0	20									
75	2									
100	2									

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4.2.3 High-impact shock test. The heater shall be subjected to the high-impact mechanical shock requirements for grade A, class I of MIL-S-901.

4.2.4 Vibration test. The heater shall be vibration tested in accordance with the requirements of type I of MIL-STD-167 with the heater operating at approximately 25 percent of full load.

4.2.5 Examination. The heater shall be thoroughly examined to ascertain that the material, workmanship, and design are in conformance with this specification.

4.3 Sampling for production inspection.

4.3.1 Lot. All heaters of the same type, class and capacity offered for delivery at one time shall be considered a lot.

4.3.2 Sampling for examination. A random sample of heaters shall be selected from each lot in accordance with MIL-STD-105 at inspection level II for the examination of 4.4. The acceptable quality level (AQL) shall be 1.5 percent defective.

4.4 Examination. Each heater selected in accordance with 4.3.2 shall be thoroughly examined to ascertain that the material, workmanship, and design are in conformance with this specification.

4.5 Production test.

4.5.1 Each heater furnished under the contract or order shall be tested in accordance with 4.2.1.

4.6 Inspection of preparation for delivery. The preservation, packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.3.)

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts (see 5.3.1).

5.1.1 Heaters.

5.1.1.1 Preservation and packaging. Preservation and packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice.

5.1.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the supplier's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts and special tools.

5.1.2.1 Packaging.

5.1.2.1.1 Onboard repair parts. Non-ferrous parts such as coiled units, springs and so forth shall be unit protected and packaged in accordance with method III of MIL-P-116. All openings in piping units, and the like, shall be plugged or sealed to prevent entrance of foreign materials. Gaskets shall be packaged in accordance with method IA, sub-method IA-8 of MIL-P-116 at the option of the supplier. Gasket shall be packaged flat and provided with fiberboard stiffeners to prevent deformation. Parts shall be individually packaged except when used in sets.

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5.1.2.1.2 Tools. Tools shall be preserved and packaged in accordance with PPP-P-40.

5.1.2.2 Packing.

5.1.2.2.1 Onboard repair parts. Onboard repair parts shall be packed in accordance with 5.1.1.2.

5.1.2.2.2 Tools. Tools shall be packed in accordance with PPP-P-40.

5.1.2.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.2 Domestic shipment and storage or overseas shipment. The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 5.3.2 and 6.2).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Preservation and packaging.

5.2.1.1.1 Level A. The heaters shall be preserved and individually packaged by method III of MIL-P-116. All openings to the heater shall be sealed to prevent entrance of foreign material.

5.2.1.2 Packing.

5.2.1.2.1 Level A. Heaters, packaged as specified, shall be individually packed in boxes conforming to one of the following specifications at the option of the supplier:

Specification	Type or class
PPP-B-601	Overseas type
PPP-B-621	Class 2

Boxes shall be modified by the addition of skids in accordance with the applicable box specifications. Cushioning, anchoring, blocking and bracing shall be in accordance with MIL-STD-1186. Box closure and strapping shall be in accordance with the applicable box specification or appendix thereto.

5.2.1.2.2 Level B. Heaters, packaged as specified, shall be individually packed in boxes conforming to one of the following specifications at the option of the supplier:

Specification	Type or class
PPP-B-601	Domestic type
PPP-B-621	Class 1

Boxes shall be modified by the addition of skids in accordance with the applicable box specifications. Cushioning, anchoring, blocking or bracing shall be in accordance with MIL-STD-1186. Box closure shall be as specified in the applicable box specification or appendix thereto.

5.2.1.3 Marking. In addition to any special marking required in the contract or order, marking of packages and shipping containers shall be in accordance with MIL-STD-129.

5.2.2 Manuals. Manuals shall be packaged, packed and marked in accordance with MIL-M-15071 for the level of shipment specified.

5.2.3 Drawings. Drawings shall be packaged, packed and marked in accordance with MIL-D-1000/2 for the level of shipment specified.)

5.3 Use of polystyrene (loose-fill) material.

5.3.1 For domestic shipment and early equipment installation and level C packaging and packing. Unless otherwise approved by the procuring activity (see 6.2), use of polystyrene (loose-fill) material for domestic shipment and early equipment installation and level C packaging and packing applications such as cushioning, filler and dunnage is prohibited. When approved, unit packages and containers (interior and exterior) shall be

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marked and labelled as follows:

"CAUTION

Contents cushioned etc. with polystyrene (loose-fill) material.
Not to be taken aboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber, fiberboard or transparent flexible cellular material."

5.3.2 For level A packaging and level A and B packing. Use of polystyrene (loose-fill) material is prohibited for level A packaging and level A and B packing applications such as cushioning, filler and dunnage.

6. NOTES

6.1 Intended use. Non-storage type water heaters are to be used where continuous hot water generation is required, space is at a minimum, low weight is required, and low steam consumption is desirable.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type (where specific type is required) and class (see 1.2.1).
(c) Capacity of heater (see 1.2.2).
(d) Whether thermometer, automatic condensate drainage components, and filtration equipment is required to be furnished as part of the heater (see 3.4.3).
(e) Electrical requirements (see 3.4.4).
(f) Shut off pressure of the supply water pumps (see 3.5.11).
(g) Operating conditions, if other than specified, and capacity required under those conditions (see 3.6.1). In particular, where water supply pressure falls outside the 40 ± 10 psig range specified in 3.6.1(c), this should be specified.
(h) Setting of hot water discharge temperature (see 3.6.4).
(i) Manuals (quantity and distribution) (see 3.9).
(j) Number of sets of special tools required if other than specified in 3.10.
(k) Preservation, packaging, packing, and marking requirements other than those required by 5.1 (see 5.2).
(l) When polystyrene "loose-fill" material is approved (see 5.3.1).
(m) Drain system backpressure.

6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.4 In order to limit the range of heaters to five basic unit sizes, ten flow ratings are used. That is, the capacity ratings of class B are different from class A to reflect the effect of temperature rise. For example, a basic heater (i.e. same coils, steam value, etc.) rated at 10 gpm for class A application (140°F. maximum setting) could be expected to deliver only about 7.7 gpm when set up for class B use (180°F. maximum setting).

6.5 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE 0101-807-2000

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